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Do Vision Transformers See Like Convolutional Neural Networks?

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Abstract

Convolutional neural networks (CNNs) have so far been the de-facto model for visual data. Recent work has shown that (Vision) Transformer models (ViT) can achieve comparable or even superior performance on image classification tasks. This raises a central question: *how are Vision Transformers solving these tasks*? Are they acting like convolutional networks, or learning entirely different visual representations? Analyzing the internal representation structure of ViTs and CNNs on image classification benchmarks, we find striking differences between the two architectures, such as ViT having more uniform representations across all layers. We explore how these differences arise, finding crucial roles played by self-attention, which enables early aggregation of global information, and ViT residual connections, which strongly propagate features from lower to higher layers. We study the ramifications for spatial localization, demonstrating ViTs successfully preserve input spatial information, with noticeable effects from different classification methods. Finally, we study the effect of (pretraining) dataset scale on intermediate features and transfer learning, and conclude with a discussion on connections to new architectures such as the MLP-Mixer.

1 Introduction

Over the past several years, the successes of deep learning on visual tasks has critically relied on convolutional neural networks [20, 16]. This is largely due to the powerful inductive bias of spatial equivariance encoded by convolutional layers, which have been key to learning general purpose visual representations for easy transfer and strong performance. Remarkably however, recent work has demonstrated that Transformer neural networks are capable of equal or superior performance on image classification tasks at large scale [14]. These Vision Transformers (ViT) operate almost *identically* to Transformers used in language [13], using self-attention, rather than convolution, to aggregate information across locations. This is in contrast with a large body of prior work, which has focused on more explicitly incorporating image-specific inductive biases [30, 9, 4]

This breakthrough highlights a fundamental question: *how* are Vision Transformers solving these image based tasks? Do they act like convolutions, learning the same inductive biases from scratch? Or are they developing novel task representations? What is the role of scale in learning these representations? And are there ramifications for downstream tasks? In this paper, we study these questions, uncovering key representational differences between ViTs and CNNs, the ways in which these difference arise, and effects on classification and transfer learning. Specifically, our contributions are:

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- We investigate the internal representation structure of ViTs and CNNs, finding striking differences between the two models, such as ViT having more uniform representations, with greater similarity between lower and higher layers.
- Analyzing how local/global spatial information is utilised, we find ViT incorporates more global information than ResNet at lower layers, leading to quantitatively different features.
- Nevertheless, we find that incorporating local information at lower layers remains vital, with large-scale pre-training data helping early attention layers learn to do this
- We study the uniform internal structure of ViT, finding that skip connections in ViT are even more influential than in ResNets, having strong effects on performance and representation similarity.
- Motivated by potential future uses in object detection, we examine how well input spatial information is preserved, finding connections between spatial localization and methods of classification.
- We study the effects of dataset scale on transfer learning, with a linear probes study revealing its importance for high quality intermediate representations.

2 Related Work

Developing non-convolutional neural networks to tackle computer vision tasks, particularly Transformer neural networks [44] has been an active area of research. Prior works have looked at *local* multiheaded self-attention, drawing from the structure of convolutional receptive fields [30, 36], directly combining CNNs with self-attention [4, 2, 46] or applying Transformers to smaller-size images [6, 9]. In comparison to these, the Vision Transformer [14] performs even less modification to the Transformer architecture, making it especially interesting to compare to CNNs. Since its development, there has also been very recent work analyzing aspects of ViT, particularly robustness [3, 31, 28] and effects of self-supervision [5, 7]. Other recent related work has looked at designing hybrid ViT-CNN models [49, 11], drawing on structural differences between the models. Comparison between Transformers and CNNs are also recently studied in the text domain [41].

Our work focuses on the representational structure of ViTs. To study ViT representations, we draw on techniques from neural network representation similarity, which allow the quantitative comparisons of representations within and across neural networks [17, 34, 26, 19]. These techniques have been very successful in providing insights on properties of different vision architectures [29, 22, 18], representation structure in language models [48, 25, 47, 21], dynamics of training methods [33, 24] and domain specific model behavior [27, 35, 38]. We also apply *linear probes* in our study, which has been shown to be useful to analyze the learned representations in both vision [1] and text [8, 32, 45] models.

3 Background and Experimental Setup

Our goal is to understand whether there are differences in the way ViTs represent and solve image tasks compared to CNNs. Based on the results of Dosovitskiy et al. [14], we take a representative set of CNN and ViT models — ResNet50x1, ResNet152x2, ViT-B/32, ViT-B/16, ViT-L/16 and ViT-H/14. Unless otherwise specified, models are trained on the JFT-300M dataset [40], although we also investigate models trained on the ImageNet ILSVRC 2012 dataset [12, 37] and standard transfer learning benchmarks [50, 14]. We use a variety of analysis methods to study the layer representations of these models, gaining many insights into how these models function. We provide further details of the experimental setting in Appendix A.

Representation Similarity and CKA (Centered Kernel Alignment): Analyzing (hidden) layer representations of neural networks is challenging because their features are distributed across a large number of neurons. This distributed aspect also makes it difficult to meaningfully compare representations across neural networks. Centered kernel alignment (CKA) [17, 10] addresses these challenges, enabling quantitative comparisons of representations within and across networks. Specifically, CKA takes as input $\mathbf{X} \in \mathbb{R}^{m \times p_1}$ and $\mathbf{Y} \in \mathbb{R}^{m \times p_2}$ which are representations (activation matrices), of two layers, with p_1 and p_2 neurons respectively, evaluated on the same m examples. Letting $\boldsymbol{K} = \boldsymbol{X}\boldsymbol{X}^\top$ and $\boldsymbol{L} = \boldsymbol{Y}\boldsymbol{Y}^\top$ denote the Gram matrices for the two layers (which measures the similarity of a pair of datapoints according to layer representations) CKA computes:

$$\mathrm{CKA}(\boldsymbol{K}, \boldsymbol{L}) = \frac{\mathrm{HSIC}(\boldsymbol{K}, \boldsymbol{L})}{\sqrt{\mathrm{HSIC}(\boldsymbol{K}, \boldsymbol{K})\mathrm{HSIC}(\boldsymbol{L}, \boldsymbol{L})}}, \tag{1}$$

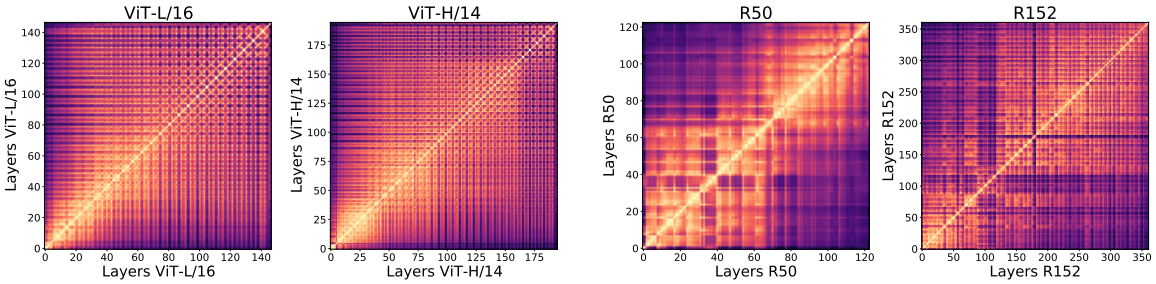

Figure 1: Representation structure of ViTs and convolutional networks show significant differences, with ViTs having highly similar representations throughout the model, while the ResNet models show much lower similarity between lower and higher layers. We plot CKA similarities between all pairs of layers across different model architectures. The results are shown as a heatmap, with the x and y axes indexing the layers from input to output. We observe that ViTs have relatively uniform layer similarity structure, with a clear grid-like pattern and large similarity between lower and higher layers. By contrast, the ResNet models show clear stages in similarity structure, with smaller similarity scores between lower and higher layers.

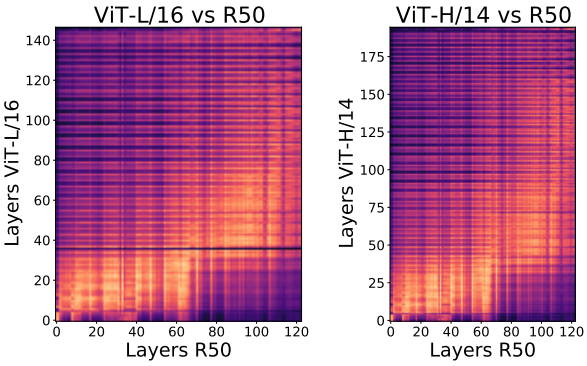

Figure 2: Cross model CKA heatmap between ViT and ResNet illustrate that a larger number of lower layers in the ResNet are similar to a smaller set of the lowest ViT layers. We compute a CKA heatmap comparing all layers of ViT to all layers of ResNet, for two different ViT models. We observe that the lower half of ResNet layers are similar to around the lowest quarter of ViT layers. The remaining half of the ResNet is similar to approximately the next third of ViT layers, with the highest ViT layers dissimilar to lower and higher ResNet layers.

where HSIC is the Hilbert-Schmidt independence criterion [15]. Given the centering matrix $\boldsymbol{H} = \boldsymbol{I}_n - \frac{1}{n}\mathbf{1}\mathbf{1}^\mathsf{T}$ and the centered Gram matrices $\boldsymbol{K}' = \boldsymbol{H}\boldsymbol{K}\boldsymbol{H}$ and $\boldsymbol{L}' = \boldsymbol{H}\boldsymbol{L}\boldsymbol{H}$, $\mathrm{HSIC}(\boldsymbol{K}, \boldsymbol{L}) = \mathrm{vec}(\boldsymbol{K}') \cdot \mathrm{vec}(\boldsymbol{L}')/(m-1)^2$, the similarity between these centered Gram matrices. CKA is invariant to orthogonal transformation of representations (including permutation of neurons), and the normalization term ensures invariance to isotropic scaling. These properties enable meaningful comparison and analysis of neural network hidden representations. To work at scale with our models and tasks, we approximate the unbiased estimator of HSIC [39] using minibatches, as suggested in [29].

4 Representation Structure of ViTs and Convolutional Networks

We begin our investigation by using CKA to study the internal representation structure of each model. How are representations propagated within the two architectures, and are there signs of functional differences? To answer these questions, we take every pair of layers $\boldsymbol{X}, \boldsymbol{Y}$ within a model and compute their CKA similarity. Note that we take representations not only from outputs of ViT/ResNet blocks, but also from intermediate layers, such as normalization layers and the hidden activations inside a ViT MLP. Figure 1 shows the results as a heatmap, for multiple ViTs and ResNets. We observe clear differences between the internal representation structure between the two model architectures: (1) ViTs show a much more uniform similarity structure, with a clear grid like structure (2) lower and higher layers in ViT show much greater similarity than in the ResNet, where similarity is divided into different (lower/higher) stages.

We also perform cross-model comparisons, where we take all layers $\boldsymbol{X}$ from ViT and compare to all layers $\boldsymbol{Y}$ from ResNet. We observe (Figure 2) that the lower half of 60 ResNet layers are similar to approximately the lowest quarter of ViT layers. In particular, many more lower layers in the ResNet are needed to compute similar representations to the lower layers of ViT. The top half of the ResNet is approximately similar to the next third of the ViT layers. The final third of ViT layers is less similar to